ITDLUPanel - Presentation Documentation

Äáº¡i há»c ÄÃ Láº¡t | Â© 2026

PHáº¦N 1: GIá»šI THIá»†U CHUNG (v1.0)

ITDLUPanel - Báº£ng Äiá»u Khiá»ƒn Quáº£n LÃ½ MÃ¡y Chá»§ Linux

Äáº¡i há»c ÄÃ Láº¡t | Â© 2026

Slide 1: Giá»›i Thiá»‡u ITDLUPanel

ITDLUPanel

Báº£ng Äiá»u Khiá»ƒn Quáº£n LÃ½ MÃ¡y Chá»§ Linux

Single Binary â€¢ Zero Dependencies â€¢ Production-Ready

ÄÆ°á»£c phÃ¡t triá»ƒn bá»Ÿi **Äáº¡i há»c ÄÃ Láº¡t**

Slide 2: Tá»•ng Quan

ITDLUPanel LÃ GÃ¬?

ITDLUPanel (IT Äáº¡i há»c ÄÃ Láº¡t Panel) lÃ má»™t **báº£ng Ä‘iá»u khiá»ƒn quáº£n lÃ½ mÃ¡y chá»§ Linux hiá»‡n Ä‘áº¡i**, Ä‘Ã³ng gÃ³i trong má»™t file nhá»‹ phÃ¢n duy nháº¥t (~60MB) vá»›i **khÃ´ng phá»¥ thuá»™c runtime bÃªn ngoÃ i**.

Äiá»ƒm Ná»•i Báºt

- **Single Binary**: ToÃ n bá»™ stack trong má»™t file Go
- **Zero Dependencies**: KhÃ´ng cáº§n cÃ i Ä‘áº·t Node.js, Python hay database riÃªng
- **Cross-Platform**: BiÃªn dá»‹ch khÃ´ng cáº§n CGO, cháº¡y trÃªn má»i Linux
- **Production-Ready**: ÄÃ£ Ä‘Æ°á»£c kiá»ƒm chá»©ng táº¡i mÃ´i trÆ°á»ng thá»±c táº¿

Slide 3: Táº¡i Sao Chá»n ITDLUPanel?

VÃ¬ Sao Chá»n ITDLUPanel?

So SÃ¡nh Vá»›i CÃ¡c Giáº£i PhÃ¡p KhÃ¡c

TiÃªu ChÃ	ITDLUPanel	cPanel	DirectAdmin
Chi phÃ	Miá»…n phÃ	Tráº£ phÃ	Tráº£ phÃ
Dependencies	KhÃ´ng	Cáº§n nhiá»u	Cáº§n nhiá»u
Giao diá»‡n	Hiá»‡n Ä‘áº¡i (React)	CÅ©	Trung bÃ¬nh
Docker tÃch há»£p	âœ…	âŒ	âŒ
WebSocket Terminal	âœ…	âŒ	âŒ

PhÃ¹ Há»£p Cho

- **TrÆ°á»ng há»c & Ä‘áº¡i há»c**: Quáº£n lÃ½ server há»c táºp
- **Doanh nghiá»‡p nhá»**: Host website ná»™i bá»™
- **DevOps**: MÃ´i trÆ°á»ng development/staging

Slide 4: Technology Stack

CÃ´ng Nghá»‡ Sá» Dá»¥ng

Backend - Go 1.25

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â”‚  Gin Framework          â”‚  JWT Auth         â”‚  SQLite â”‚
â”‚  WebSocket             â”‚  TOTP 2FA         â”‚  gopsutilâ”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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Component	Technology	Version
Language	Go	1.25
HTTP Framework	Gin	1.12
Database	SQLite	modernc.org/sqlite 1.47
Auth	JWT	golang-jwt/jwt 5.3.1
Terminal	xterm.js PTY	creack/pty 1.1.24
Storage	MinIO Go	7.0.100

Frontend - React 19

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â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚  React 19              â”‚  Ant Design 6.3      â”‚  Vite  â”‚
â”‚  TypeScript 5.9        â”‚  Tailwind CSS 4      â”‚  i18n  â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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Component	Technology	Version
Framework	React	19.2.4
UI Library	Ant Design	6.3.3
Build Tool	Vite	8.0.1
State	Zustand	5.0.12
Charts	Recharts	3.8.0
Terminal	xterm.js	6.0.0

Slide 5: Kiáº¿n TrÃºc Há»‡ Thá»‘ng

Kiáº¿n TrÃºc Tá»•ng Quan

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â”‚                     Browser (React 19 SPA)                      â”‚
â”‚         Ant Design 6 â€¢ xterm.js â€¢ Monaco Editor                â”‚
â”‚         Recharts â€¢ Zustand â€¢ React Query â€¢ i18n                â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¬â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¬â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
                 â”‚  REST API (~160)       â”‚  WebSocket (2)
                 â”‚  HTTPS / JWT          â”‚  PTY Terminal + Monitor
                 â–¼                        â–¼
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚              Single Go Binary (go:embed)                       â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”‚   Gin   â”‚â†’ â”‚  24 Handlers â”‚â†’ â”‚  33+ Services         â”‚   â”‚
â”‚  â”‚  Router â”‚  â”‚  (REST API)  â”‚  â”‚  14 Installers        â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â”‚  4 Vhost Generators  â”‚   â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¬â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â”‚  â”‚   WS    â”‚â†’ â”‚  Terminal    â”‚             â”‚              â”‚
â”‚  â”‚   Hub   â”‚  â”‚  Monitor     â”‚            â–¼              â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”    â”‚  SQLite (25 tables)  â”‚   â”‚
â”‚  â”‚  Middleware Stack (11)   â”‚    â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
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â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¬â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
                             â–¼
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                     Linux Host (Ubuntu/Debian)                 â”‚
â”‚   Nginx/Apache/OLS â€¢ PHP-FPM â€¢ MySQL/MariaDB â€¢ Redis       â”‚
â”‚   Docker â€¢ Fail2ban â€¢ PureFTPd â€¢ Certbot                     â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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Slide 6: CÃ¡c Module ChÃnh

TÃnh NÄƒng ChÃnh

Module	MÃ´ Táº£
ðŸ“Š **Dashboard**	Theo dÃµi CPU, RAM, Disk, Network realtime
ðŸŒ **Websites**	Quáº£n lÃ½ site, domain, PHP version, Nginx/Apache
ðŸ—„ï¸ **Databases**	MySQL/MariaDB - create, backup, restore
ðŸ“ **File Manager**	TrÃ¬nh duyá»‡t file vá»›i Monaco Editor
ðŸ’» **Terminal**	WebSocket terminal vá»›i multi-tab
ðŸ”’ **Security**	Firewall, IP blocking, Fail2ban
ðŸ³ **Docker**	38+ one-click app templates
â° **Cron Jobs**	30 templates, schedule builder
ðŸ“ **Logs**	Panel, SSH, website, audit logs
ðŸ” **SSL**	Let's Encrypt auto-issue, auto-renewal
ðŸ’¾ **Backups**	FTP, S3, Google Drive, OneDrive
ðŸ‘¥ **Users**	RBAC - Admin/Operator/Viewer + 2FA

Slide 7: Dashboard - GiÃ¡m SÃ¡t Realtime

Dashboard ThÃ´ng Minh

Trá»±c Quan HÃ³a Dá»¯ Liá»‡u

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â”‚  CPU Usage          RAM Usage         Disk Usage       â”‚
â”‚  â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–‘â–‘ 78%    â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–‘â–‘â–‘â–‘ 62%   â–ˆâ–ˆâ–ˆâ–‘â–‘â–‘â–‘â–‘â–‘ 31%     â”‚
â”‚  [==========]       [==========]       [==========]    â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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TÃnh NÄƒng Dashboard

- âœ… **Realtime Monitoring**: Cáºp nháºt má»—i 2 giÃ¢y qua WebSocket
- âœ… **Interactive Charts**: Biá»ƒu Ä‘á»“ tÆ°Æ¡ng tÃ¡c vá»›i Recharts
- âœ… **Network Stats**: Bandwidth in/out theo thá»i gian thá»±c
- âœ… **Process List**: Top processes Ä‘ang cháº¡y
- âœ… **Service Status**: Health check cÃ¡c dá»‹ch vá»¥

CÃ¡c Chá»‰ Sá»‘ Theo DÃµi

Metric	Update Interval	Retention
CPU	2s	1 hour
Memory	2s	1 hour
Disk I/O	5s	1 hour
Network	2s	1 hour

Slide 8: Quáº£n LÃ½ Websites

Quáº£n LÃ½ Website ChuyÃªn Nghiá»‡p

Há»— Trá»£ Äa Web Server

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â”‚  Web Server Support                                     â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”                  â”‚
â”‚  â”‚  Nginx  â”‚  â”‚ Apache  â”‚  â”‚   OLS   â”‚                  â”‚
â”‚  â”‚   âœ…    â”‚  â”‚   âœ…    â”‚  â”‚   âœ…    â”‚                  â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜                  â”‚
â”‚                                                         â”‚
â”‚  PHP Versions: 7.4 â€¢ 8.0 â€¢ 8.1 â€¢ 8.2 â€¢ 8.3 â€¢ 8.4     â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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TÃnh NÄƒng Ná»•i Báºt

- ðŸ”„ **PHP Version Switching**: Chuyá»ƒn Ä‘á»•i PHP version dá»… dÃ ng
- ðŸ“ **Config Editor**: Chá»‰nh sá»a Nginx/Apache config
- ðŸŒ **Domain Aliases**: Quáº£n lÃ½ multiple domains
- ðŸ“Š **Access Logs**: Xem logs theo thá»i gian thá»±c
- ðŸ”’ **SSL Integration**: TÃch há»£p Let's Encrypt

Quick Actions

```
[Create Site]  [PHP Settings]  [SSL Cert]  [Config Editor]
```

Slide 9: Quáº£n LÃ½ Databases

Quáº£n LÃ½ Database ToÃ n Diá»‡n

Supported Databases

Database	Port	Features
MySQL	3306	Full support
MariaDB	3306	Full support
Redis	6379	Cache management

Database Features

- âœ… **Create/Drop**: Táº¡o vÃ xÃ³a database nhanh chÃ³ng
- âœ… **User Management**: PhÃ¢n quyá»n user database
- âœ… **Backup/Restore**: Sao lÆ°u vá»›i nhiá»u destination
- âœ… **phpMyAdmin**: Giao diá»‡n web quáº£n lÃ½
- âœ… **Connection Info**: ThÃ´ng tin káº¿t ná»‘i chi tiáº¿t

Backup Destinations

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚  ðŸ“ FTP Server    â”‚   â˜ï¸ S3/MinIO   â”‚   ðŸ”µ Google Drive â”‚
â”‚  ðŸ“ OneDrive      â”‚   ðŸ“ Local      â”‚   ðŸ“ SFTP          â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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Slide 10: File Manager & Code Editor

File Manager ThÃ´ng Minh

Monaco Editor TÃch Há»£p

- ðŸ”¥ **Syntax Highlighting**: Há»— trá»£ 50+ ngÃ´n ngá»¯
- ðŸ“‚ **File Tree**: Duyá»‡t file vá»›i tree view

- ðŸ“ **Compress/Extract**: ZIP, TAR, GZ
- ðŸ” **Permission Editor**: Chmod trá»±c quan
- â **Bookmarks**: ÄÃ¡nh dáº¥u thÆ° má»¥c yÃªu thÃch
- ðŸ” **Search**: TÃ¬m kiáº¿m file nhanh

Supported File Types

```
Languages: JavaScript, TypeScript, Python, Go, PHP,
           Ruby, Java, C++, Rust, HTML, CSS, JSON, YAML...
```

Quick Actions

Action	Description
New File	Táº¡o file má»›i
Upload	Upload file lÃªn server
Download	Táº£i file vá» local
Compress	NÃ©n file/folder
Extract	Giáº£i nÃ©n archive

Slide 11: WebSocket Terminal

Terminal Trong TrÃ¬nh Duyá»‡t

TÃnh NÄƒng Terminal

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â”‚  ITDLUPanel Terminal                           [+] [_] [X]   â”‚
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â”‚  â”‚  admin@server:~$ ssh user@server                       â”‚ â”‚
â”‚  â”‚  user@server's password: ********                      â”‚ â”‚
â”‚  â”‚  Welcome to Ubuntu 22.04 LTS                           â”‚ â”‚
â”‚  â”‚  user@server:~$ ls -la                                  â”‚ â”‚
â”‚  â”‚  drwxr-xr-x  5 admin admin  4096 Jun 29 10:00 .       â”‚ â”‚
â”‚  â”‚  drwxr-xr-x 14 admin admin  4096 Jun 29 09:30 ..      â”‚ â”‚
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â”‚  [Tab 1] [Tab 2] [+]              [Fullscreen] [Theme: Dark] â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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- ðŸ’» **Full PTY Support**: Terminal Ä‘áº§y Ä‘á»§ chá»©c nÄƒng
- ðŸ“‘ **Multi-Tab**: Nhiá»u terminal cÃ¹ng lÃºc
- ðŸ”² **Fullscreen**: Cháº¿ Ä‘á»™ toÃ n mÃ n hÃ¬nh
- ðŸŽ¨ **Theme Support**: Light/Dark theme
- ðŸ“œ **Scrollback**: Buffer khÃ´ng giá»›i háº¡n

Technology

- **Backend**: gorill/websocket + creack/pty
- **Frontend**: xterm.js vá»›i custom themes

Slide 12: Docker Management

Docker Container Management

Container Features

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â”‚  ðŸ³ Docker Status: Running                               â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”        â”‚
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â”‚  â”‚ â— RUN   â”‚ â”‚ â— RUN   â”‚ â”‚ â— RUN   â”‚ â”‚ â— RUN   â”‚        â”‚
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Management Capabilities

Feature	Description
Containers	Start, stop, restart, remove
Images	Pull, build, remove images
Networks	Create and manage networks
Volumes	Persistent storage management
Compose	Docker Compose editor & runner

One-Click App Templates (38+)

```
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â”‚  Popular Templates                      â”‚
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â”‚  â”‚ WordPress â”‚ â”‚  Nextcloudâ”‚ â”‚  GitLab  â”‚              â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜              â”‚
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â”‚  â”‚  Matomo   â”‚ â”‚   Jira   â”‚ â”‚ Prometheusâ”‚              â”‚
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Slide 13: Security Features

Báº£o Máºt ToÃ n Diá»‡n

Security Stack

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â”‚                  Security Layers                        â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”‚
â”‚  â”‚  JWT Auth (15min access + 7day refresh token)     â”‚ â”‚
â”‚  â”‚  TOTP 2FA (RFC 6238)                              â”‚ â”‚
â”‚  â”‚  RBAC (Admin/Operator/Viewer)                     â”‚ â”‚
â”‚  â”‚  bcrypt password hashing (cost=12)                â”‚ â”‚
â”‚  â”‚  CSRF Protection                                   â”‚ â”‚
â”‚  â”‚  Rate Limiting (300 req/min per IP)               â”‚ â”‚
â”‚  â”‚  Audit Logging                                     â”‚ â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”‚
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Security Features

Feature	Description
ðŸ”¥ **Firewall**	UFW rules management
ðŸš« **IP Blocking**	Block malicious IPs
ðŸ›¡ï¸ **Fail2ban**	Auto-ban failed login attempts
ðŸ”‘ **SSH Config**	SSH hardening options
ðŸ“Š **Port Monitor**	Track open ports
ðŸ“ **Audit Logs**	Log all admin actions

RBAC Roles

Role	Permissions
Admin	Full access
Operator	Manage services, no user management
Viewer	Read-only access

Slide 14: Há»‡ Thá»‘ng Logs ToÃ n Diá»‡n

Quáº£n LÃ½ Logs ThÃ´ng Minh

Tá»•ng Quan Logs

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â”‚  ðŸ“‹ ITDLUPanel Log Center                              â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”‚
â”‚  â”‚  ðŸ” Filter: [All Logs â–¼] [Date Range] [Search...]           â”‚ â”‚
â”‚  â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤ â”‚
â”‚  â”‚  ðŸ“… 2026-06-30 14:25:03  [AUDIT]  User admin logged in       â”‚ â”‚
â”‚  â”‚  ðŸ“… 2026-06-30 14:24:58  [ERROR]  Nginx: connection failed   â”‚ â”‚
â”‚  â”‚  ðŸ“… 2026-06-30 14:23:15  [INFO]   Backup completed: db.sql   â”‚ â”‚
â”‚  â”‚  ðŸ“… 2026-06-30 14:22:01  [WARN]   SSH: failed login attempt  â”‚ â”‚
â”‚  â”‚  ðŸ“… 2026-06-30 14:20:33  [PANEL]  Service nginx restarted    â”‚ â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”‚
â”‚  [Download] [Export] [Clear Old] [Settings]                        â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

CÃ¡c Loáº¡i Logs

|| Type | Description | Location | |------|-------------|----------| | ðŸ“Š **Panel Logs** | Hoáº¡t Ä‘á»™ng cá»§a panel (login, logout, config changes) | `/var/log/itdlupanel/panel.log` | | ðŸ’» **SSH Logs** | Káº¿t ná»‘i SSH, failed login attempts | `/var/log/auth.log` | | ðŸŒ **Website Logs** | Access logs, error logs per domain | `/var/www/{domain}/logs/` | | ðŸ”’ **Audit Logs** | Táº¥t cáº£ actions cá»§a admin | SQLite database | | ðŸ³ **Docker Logs** | Container logs, stdout/stderr | `docker logs {container}` | | ðŸ”„ **Service Logs** | Systemd services, cron jobs | `/var/log/syslog` |

TÃnh NÄƒng ChÃnh

- ðŸ” **Advanced Filtering**: Lá»c theo loáº¡i, ngÃ y, má»©c Ä‘á»™ nghiÃªm trá»ng
- ðŸ“… **Date Range Picker**: Chá»n khoáº£ng thá»i gian xem logs
- ðŸ”Ž **Full-Text Search**: TÃ¬m kiáº¿m ná»™i dung log
- ðŸ“¥ **Export**: Xuáº¥t logs ra JSON, CSV, TXT
- ðŸ—œï¸ **Auto-Compression**: NÃ©n logs cÅ© tá»± Ä‘á»™ng
- â° **Retention Policy**: Cáº¥u hÃ¬nh sá»‘ ngÃ y lÆ°u trá»¯

Log Levels

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚  Log Levels                                                        â”‚
â”‚                                                                   â”‚
â”‚  ðŸ”´ CRITICAL  - Lá»—i nghiÃªm trá»ng, cáº§n can thiá»‡p ngay          â”‚
â”‚  ðŸŸ  ERROR     - Lá»—i hoáº¡t Ä‘á»™ng khÃ´ng bÃ¬nh thÆ°á»ng                â”‚
â”‚  ðŸŸ¡ WARNING   - Cáº£nh bÃ¡o, cÃ³ thá»ƒ áº£nh hÆ°á»Ÿng                    â”‚
â”‚  ðŸ”µ INFO      - ThÃ´ng tin hoáº¡t Ä‘á»™ng bÃ¬nh thÆ°á»ng                â”‚
â”‚  âšª DEBUG     - Chi tiáº¿t debug (dev mode)                      â”‚
â”‚                                                                   â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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Security Audit Logs

|| Field | Description | |-------|-------------| | Timestamp | Thá»i gian chÃnh xÃ¡c (timezone) | | User | TÃ i khoáº£n thá»±c hiá»‡n | | Action | HÃ nh Ä‘á»™ng (create/delete/update/restart) | | Resource | Äá»‘i tÆ°á»£ng bá»‹ tÃ¡c Ä‘á»™ng | | IP Address | Äá»‹a chá»‰ IP cá»§a ngÆ°á»i dÃ¹ng | | Result | ThÃ nh cÃ´ng/Tháº¥t báº¡i |

Website Access Logs

```
# Apache/Nginx Access Log Format
[30/Jun/2026:14:25:03 +0700] "GET /api/users HTTP/1.1" 200 1234
[30/Jun/2026:14:25:04 +0700] "POST /login HTTP/1.1" 302 0
[30/Jun/2026:14:25:05 +0700] "GET /dashboard HTTP/1.1" 200 5678
```

Quick Actions

|| Action | Description | |--------|-------------| | View Live | Xem logs realtime vá»›i auto-refresh | | Download | Táº£i toÃ n bá»™ log file | | Search | TÃ¬m kiáº¿m ná»™i dung | | Archive | NÃ©n vÃ lÆ°u trá»¯ logs cÅ© | | Purge | XÃ³a logs theo retention policy |

Slide 15: SSL Certificates

Quáº£n LÃ½ SSL Certificate

SSL Features

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â”‚  ðŸ”’ SSL Certificate Manager                              â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”‚
â”‚  â”‚  Domain: example.com                                â”‚ â”‚
â”‚  â”‚  Provider: Let's Encrypt                            â”‚ â”‚
â”‚  â”‚  Valid Until: 2026-09-29                            â”‚ â”‚
â”‚  â”‚  Auto-Renewal: âœ… Enabled                           â”‚ â”‚
â”‚  â”‚  [View] [Renew] [Download] [Delete]                â”‚ â”‚
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Certificate Options

Type	Description
Let's Encrypt	Free auto-renewal SSL
Self-Signed	For internal/testing
Custom	Upload your own cert

Features

- âœ… **Auto-Issue**: Tá»± Ä‘á»™ng cáº¥p phÃ¡t certificate
- âœ… **Auto-Renewal**: Gia háº¡n tá»± Ä‘á»™ng trÆ°á»›c 30 ngÃ y
- âœ… **Wildcard Support**: Há»— trá»£ wildcard certificates
- âœ… **Multi-Domain**: Nhiá»u domain trong 1 cert

Slide 16: Backup & Storage

Backup & Cloud Storage

Supported Storage Backends

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â”‚  â˜ï¸ Cloud Storage Options                              â”‚
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â”‚  â”‚  Amazon S3  â”‚  â”‚  Google     â”‚  â”‚  OneDrive   â”‚    â”‚
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â”‚  â”‚  (S3-like)  â”‚  â”‚   Server    â”‚  â”‚   Storage   â”‚    â”‚
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```

Backup Features

- âœ… **Scheduled Backups**: Äáº·t lá»‹ch backup tá»± Ä‘á»™ng
- âœ… **Incremental Backup**: Chá»‰ backup thay Ä‘á»•i
- âœ… **Compression**: NÃ©n gzip/tar
- âœ… **Encryption**: MÃ£ hÃ³a backup files
- âœ… **Retention Policy**: XÃ³a backup cÅ© tá»± Ä‘á»™ng

OAuth2 Integration

- Google Drive via OAuth2
- OneDrive via OAuth2
- S3/MinIO via Access Keys

Slide 17: Internationalization

Äa NgÃ´n Ngá»¯

Supported Languages

```
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â”‚  â”‚   (EN)      â”‚              â”‚    (VI)     â”‚         â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜              â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜         â”‚
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```

i18n Coverage

Component	Coverage
Backend API Messages	100%
Frontend UI	100%
Error Messages	100%
Documentation	EN + VI

Implementation

- **Frontend**: i18next vá»›i react-i18next
- **Backend**: Custom i18n package
- **Dynamic Switching**: Äá»•i ngÃ´n ngá»¯ khÃ´ng cáº§n reload

---

Slide 18: Code Statistics

Dá»± Ãn Báº±ng Con Sá»‘

Codebase Metrics

Metric	Count
Go files	112 files â€¢ 43,314 LOC
TypeScript files	93 files
API endpoints	~160 REST + 2 WebSocket
Database tables	25
Docker templates	38 one-click apps
Cron templates	30 pre-built
Software installers	14
Frontend pages	17 (lazy-loaded)
Shared components	15

Backend Architecture

```
Middleware Stack: 11 layers
API Handlers: 24 files
Services: 33+
Installers: 14
Vhost Generators: 4 (Nginx, Apache, OLS, Node)
```

Frontend Architecture

```
Pages: 17 (all lazy-loaded)
Components: 15 shared
Zustand Stores: 5
```

```
Custom Hooks: 7
```

Slide 19: Deployment

Triá»ƒn Khai Dá»… DÃ ng

Quick Start

```
# 1. Build
cd web && npm install && npx vite build && cd ..
make build

# 2. Configure
cp config/panel.json.example config/panel.json
vim config/panel.json

# 3. Run
./bin/itdlupanel -config config/panel.json

# 4. Access
# Open https://localhost:8888
```

Default Credentials

Field	Value
Username	`admin`
Password	`admin@123456`
Default Port	`8888`

System Requirements

Resource	Minimum	Recommended
RAM	1 GB	2 GB
CPU	1 core	2+ cores
Disk	5 GB	20 GB
OS	Ubuntu 18.04+	Ubuntu 22.04 LTS

Slide 20: Roadmap

Lá»™ TrÃ¬nh PhÃ¡t Triá»ƒn

Phase 1: âœ… Completed

- [x] Core panel functionality
- [x] Docker management
- [x] WebSocket terminal
- [x] SSL certificates
- [x] Multi-language support

Phase 2: ðŸš§ In Progress

- [ ] Kubernetes cluster management
- [ ] Advanced monitoring (Prometheus/Grafana)
- [ ] Plugin marketplace
- [ ] API documentation (OpenAPI)

Phase 3: ðŸ“‹ Planned

- [ ] Multi-server management
- [ ] Mobile app (React Native)
- [ ] AI-powered suggestions

- [ ] GitOps integration

Slide 21: Káº¿t Luáºn

Cáº£m Æ n!

ITDLUPanel

Contact & Support

- ðŸŒ **Website**: https://github.com/itdlu/ITDLUPanel
- ðŸ“§ **Email**: support@itdlu.edu.vn
- ðŸ“– **Docs**: https://docs.itdlu.edu.vn

Key Takeaways

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚  âœ“ Single Binary Deployment                             â”‚
â”‚  âœ“ Zero External Dependencies                          â”‚
â”‚  âœ“ Production-Ready Security (JWT, 2FA, RBAC)          â”‚
â”‚  âœ“ Docker & Cloud Storage Integration                  â”‚
â”‚  âœ“ Real-time WebSocket Terminal                        â”‚
â”‚  âœ“ EN/VI Language Support                              â”‚
â”‚  âœ“ MIT License - Completely Free                       â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
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Generated by ITDLUPanel Documentation System

PHáº¦N 2: PHÃ‚N TÃCH Ká»¸ THUáº¬T CHI TIáº¾T (v2.0)

ITDLUPanel v2 - PhÃ¢n TÃch Ká»¹ Thuáºt Chi Tiáº¿t

Äáº¡i há»c ÄÃ Láº¡t | Â© 2026

Slide 1: Tá»•ng Quan Kiáº¿n TrÃºc

ITDLUPanel v2 - Technical Deep Dive

PhÃ¢n TÃch Ká»¹ Thuáºt ToÃ n Diá»‡n

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```

Key Metrics

Metric	Value
Go Files	112 files
Lines of Code	43,314 LOC
API Endpoints	~160 REST + 2 WS
Database Tables	25
Services	33+
Middleware Layers	11

Slide 2: Backend Layer Architecture

Backend Go Architecture

Layered Design Pattern

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```

Slide 3: Middleware Stack Chi Tiáº¿t

Middleware Pipeline Deep Dive

11-Layer Middleware Stack

Layer 1: Recovery Middleware

```
func RecoveryMiddleware() gin.HandlerFunc {
    return func(c *gin.Context) {
        defer func() {
            if err := recover(); err != nil {
                // Log panic with stack trace
                zerolog.Error().
                    Interface("panic", err).
                    Str("stack", string(debug.Stack())).
                    Msg("Panic recovered")

                c.AbortWithStatusJSON(500, gin.H{
                    "error": "Internal server error",
                })
            }
        }()
        c.Next()
    }
}
```

Layer 2: Structured Logging

```
// zerolog with lumberjack rotation
zerolog.TimeFieldFormat = zerolog.TimeFormatUnix
log := zerolog.New(os.Stdout).
    Level(zerolog.InfoLevel).
    With().
    Timestamp().
    Caller().
    Logger()
```

Layer 3-6: Security Layers

Layer	Header	Purpose
Security	X-Frame-Options	Clickjacking protection
Security	X-Content-Type-Options	MIME sniffing prevention
Security	X-XSS-Protection	XSS filter (legacy)
Security	Strict-Transport-Security	Force HTTPS
CORS	Access-Control-*	Cross-origin requests

Layer 7: Rate Limiting

```
Algorithm: Token Bucket
â”œâ”€â”€ Rate: 300 requests/minute
â”œâ”€â”€ Per: IP address
â””â”€â”€ Response: 429 Too Many Requests
```

Layer 8: CSRF Protection

```
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â”‚  CSRF Token Flow                                           â”‚
â”‚                                                             â”‚
â”‚  1. Server generates: csrf_token = cryptoéšæœºæ•°          â”‚
â”‚  2. Set-Cookie: HttpOnly; SameSite=Strict                â”‚
â”‚  3. Response includes: X-CSRF-Token header               â”‚
â”‚  4. Client must send both cookie + header                 â”‚
â”‚  5. Server validates: cookie == header                    â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Layer 9: Audit Logging

```
// Write operation audit trail
type AuditLog struct {
    UserID    uint      `db:"user_id"`
    Username  string    `db:"username"`
    Action    string    `db:"action"`     // CREATE, UPDATE, DELETE
    Resource  string    `db:"resource"`   // sites, databases, users
    Details   string    `db:"details"`    // JSON payload
    IPAddress string    `db:"ip_address"`
    Timestamp time.Time `db:"created_at"`
}
```

Layer 10-11: Auth + RBAC

```
// JWT Token Structure
type Claims struct {
    UserID      uint     `json:"user_id"`
    Username    string   `json:"username"`
    Role        string   `json:"role"`
    Permissions []string `json:"permissions"`
    jwt.RegisteredClaims
}

// RBAC Permission Check
func CheckPermission(permissions []string, required string) bool {
    for _, p := range permissions {
        if p == "*" || p == required {
            return true
        }
    }
    return false
}
```

Slide 4: Source Code Structure

Go Project Structure

Directory Layout

```
internal/
â”œâ”€â”€ api/
â”‚   â”œâ”€â”€ v1/                    # 24 REST handler files (9,814 LOC)
â”‚   â”‚   â”œâ”€â”€ auth.go            # Authentication endpoints
â”‚   â”‚   â”œâ”€â”€ site.go            # Website management
â”‚   â”‚   â”œâ”€â”€ database.go        # Database CRUD
â”‚   â”‚   â”œâ”€â”€ docker.go          # Docker management (46+ endpoints)
â”‚   â”‚   â”œâ”€â”€ file.go            # File browser/editor
â”‚   â”‚   â”œâ”€â”€ docker_compose.go  # Compose projects
â”‚   â”‚   â”œâ”€â”€ ssl.go             # SSL certificates
```

```
â”‚   â”‚   â”œâ”€â”€ backup.go          # Backup management
â”‚   â”‚   â”œâ”€â”€ security.go         # Firewall, SSH, Fail2ban
â”‚   â”‚   â”œâ”€â”€ cron.go            # Cron jobs
â”‚   â”‚   â”œâ”€â”€ log.go             # Log viewer
â”‚   â”‚   â”œâ”€â”€ monitor.go         # System metrics
â”‚   â”‚   â”œâ”€â”€ user.go            # User management
â”‚   â”‚   â”œâ”€â”€ role.go             # RBAC roles
â”‚   â”‚   â”œâ”€â”€ settings.go         # Panel settings
â”‚   â”‚   â”œâ”€â”€ software.go         # Software installation
â”‚   â”‚   â”œâ”€â”€ ftp.go              # FTP accounts
â”‚   â”‚   â”œâ”€â”€ ftp_storage.go     # FTP backup storage
â”‚   â”‚   â”œâ”€â”€ cloud_storage.go    # S3/GDrive/OneDrive
â”‚   â”‚   â”œâ”€â”€ redis.go           # Redis management
â”‚   â”‚   â”œâ”€â”€ task.go            # Background tasks
â”‚   â”‚   â”œâ”€â”€ notification.go     # Notifications
â”‚   â”‚   â”œâ”€â”€ plugin.go          # Plugin system
â”‚   â”‚   â”œâ”€â”€ waf.go             # WAF management
â”‚   â”‚   â”œâ”€â”€ system.go          # System info
â”‚   â”‚   â””â”€â”€ docker_cleanup.go   # Docker auto-cleanup
â”‚   â”‚
â”‚   â””â”€â”€ ws/                     # 2 WebSocket handlers
â”‚       â”œâ”€â”€ terminal.go         # PTY terminal (xterm.js â†” creack/pty)
â”‚       â””â”€â”€ monitor.go          # Real-time metrics stream
â”‚
â”œâ”€â”€ app/                        # Application bootstrap
â”‚   â”œâ”€â”€ app.go                  # Main entry, graceful shutdown
â”‚   â”œâ”€â”€ router.go               # Gin route registration
â”‚   â”œâ”€â”€ auth_middleware.go      # JWT + RBAC middleware
â”‚   â””â”€â”€ middleware.go          # CORS, rate limit, compression
â”‚
â”œâ”€â”€ service/                    # Business logic (23,437 LOC)
â”‚   â”œâ”€â”€ auth_service.go         # Login, register, 2FA
â”‚   â”œâ”€â”€ jwt.go                  # Token generation/validation
â”‚   â”œâ”€â”€ totp_service.go         # TOTP 2FA (RFC 6238)
â”‚   â”œâ”€â”€ user_service.go         # User CRUD
â”‚   â”œâ”€â”€ role_service.go         # RBAC role management
â”‚   â”œâ”€â”€ site_service.go         # Website CRUD + config
â”‚   â”œâ”€â”€ site_statistics_service.go  # Traffic statistics
â”‚   â”œâ”€â”€ database_service.go     # MySQL/MariaDB management
â”‚   â”œâ”€â”€ file_service.go         # File operations
â”‚   â”œâ”€â”€ docker_service.go       # Docker API wrapper
â”‚   â”œâ”€â”€ docker_compose_service.go   # Compose project management
â”‚   â”œâ”€â”€ docker_cleanup_service.go    # Auto-cleanup old images/containers
â”‚   â”œâ”€â”€ backup_service.go       # Backup orchestration
â”‚   â”œâ”€â”€ backup_incremental.go    # Incremental backup logic
â”‚   â”œâ”€â”€ backup_encryption.go    # AES-256 encryption
â”‚   â”œâ”€â”€ backup_upload_service.go    # Multi-destination upload
â”‚   â”œâ”€â”€ ssl_service.go          # Let's Encrypt integration
â”‚   â”œâ”€â”€ ssl_renewer.go          # Auto-renewal scheduler
â”‚   â”œâ”€â”€ cron_service.go         # Cron job management
â”‚   â”œâ”€â”€ terminal_service.go     # PTY session management
â”‚   â”œâ”€â”€ monitor_service.go      # System metrics collection
â”‚   â”œâ”€â”€ metrics_collector.go    # CPU/RAM/Disk/Network
â”‚   â”œâ”€â”€ security_service.go     # SSH, Fail2ban, hardening
â”‚   â”œâ”€â”€ firewall_service.go     # UFW rules management
â”‚   â”œâ”€â”€ software_service.go     # Software install/uninstall
â”‚   â”œâ”€â”€ settings_service.go     # Panel configuration
â”‚   â”œâ”€â”€ redis_service.go        # Redis management
â”‚   â”œâ”€â”€ ftp_service.go         # PureFTPd management
â”‚   â”œâ”€â”€ ftp_storage_service.go  # FTP storage backends
â”‚   â”œâ”€â”€ cloud_storage_service.go # S3/GDrive/OneDrive
â”‚   â”œâ”€â”€ notification_service.go # Email/Telegram/Webhook
â”‚   â”œâ”€â”€ email_service.go        # SMTP integration
â”‚   â”œâ”€â”€ telegram_service.go     # Telegram bot notifications
â”‚   â”œâ”€â”€ geo_service.go          # GeoIP lookup
â”‚   â”œâ”€â”€ log_service.go         # Log aggregation
â”‚   â”œâ”€â”€ waf_service.go         # WAF rule management
â”‚   â”œâ”€â”€ waf_log_service.go     # WAF access logs
â”‚   â”œâ”€â”€ haproxy_manager.go      # HAProxy configuration
â”‚   â”œâ”€â”€ coraza_spoa.go         # Coraza WAF SPOA integration
â”‚   â”œâ”€â”€ permission_cache.go     # RBAC permission caching
â”‚   â”œâ”€â”€ user_resource_service.go    # User quotas
â”‚   â”œâ”€â”€ site_defaults.go       # Default site configuration
â”‚   â”‚
â”‚   â”œâ”€â”€ installer/             # 14 software installers
â”‚   â”‚   â”œâ”€â”€ installer.go       # Base installer interface
```

```
â”‚   â”‚   â”œâ”€â”€ nginx.go          # Nginx installation
â”‚   â”‚   â”œâ”€â”€ apache.go         # Apache installation
â”‚   â”‚   â”œâ”€â”€ ols.go            # OpenLiteSpeed
â”‚   â”‚   â”œâ”€â”€ php.go            # PHP (multiple versions)
â”‚   â”‚   â”œâ”€â”€ mysql.go          # MySQL installation
â”‚   â”‚   â”œâ”€â”€ postgresql.go     # PostgreSQL
â”‚   â”‚   â”œâ”€â”€ mongodb.go        # MongoDB
â”‚   â”‚   â”œâ”€â”€ redis.go         # Redis
â”‚   â”‚   â”œâ”€â”€ memcached.go     # Memcached
â”‚   â”‚   â”œâ”€â”€ phpmyadmin.go     # phpMyAdmin
â”‚   â”‚   â”œâ”€â”€ pgadmin.go       # pgAdmin
â”‚   â”‚   â”œâ”€â”€ pureftpd.go       # PureFTPd
â”‚   â”‚   â”œâ”€â”€ supervisor.go     # Supervisor
â”‚   â”‚   â”œâ”€â”€ docker.go         # Docker Engine
â”‚   â”‚   â”œâ”€â”€ nodejs.go         # Node.js
â”‚   â”‚   â”œâ”€â”€ python.go         # Python
â”‚   â”‚   â”œâ”€â”€ golang.go         # Go
â”‚   â”‚   â”œâ”€â”€ java.go           # Java/Tomcat
â”‚   â”‚   â”œâ”€â”€ elasticsearch.go   # Elasticsearch
â”‚   â”‚   â”œâ”€â”€ s3_storage.go     # MinIO S3
â”‚   â”‚   â”œâ”€â”€ google_drive.go   # Google Drive OAuth
â”‚   â”‚   â”œâ”€â”€ onedrive.go       # OneDrive OAuth
â”‚   â”‚   â””â”€â”€ ftp_storage.go    # FTP storage
â”‚   â”‚
â”‚   â””â”€â”€ vhost/                 # 4 web server vhost generators
â”‚       â”œâ”€â”€ vhost.go          # Base vhost interface
â”‚       â”œâ”€â”€ nginx.go          # Nginx site config
â”‚       â”œâ”€â”€ apache.go         # Apache vhost
â”‚       â””â”€â”€ ols.go            # OpenLiteSpeed vhost
â”‚
â”œâ”€â”€ database/                   # SQLite + models
â”‚   â”œâ”€â”€ sqlite.go              # Database connection pool
â”‚   â”œâ”€â”€ models/               # Data models
â”‚   â”‚   â””â”€â”€ role.go          # RBAC models
â”‚   â””â”€â”€ migrations/           # 14 migration files
â”‚       â”œâ”€â”€ migrations.go     # Migration runner
â”‚       â”œâ”€â”€ 002_monitor_metrics.go
â”‚       â”œâ”€â”€ 003_firewall_settings.go
â”‚       â”œâ”€â”€ 004_phpmyadmin_default_port.go
â”‚       â”œâ”€â”€ 005_quota.go
â”‚       â”œâ”€â”€ 006_site_description.go
â”‚       â”œâ”€â”€ 007_permissions.go
â”‚       â”œâ”€â”€ 008_panel_settings.go
â”‚       â”œâ”€â”€ 009_ftp_storage.go
â”‚       â”œâ”€â”€ 010_cloud_storages.go
â”‚       â”œâ”€â”€ 011_user_resources.go
â”‚       â”œâ”€â”€ 012_notifications.go
â”‚       â”œâ”€â”€ 013_notification_templates.go
â”‚       â”œâ”€â”€ 014_site_type.go
â”‚       â”œâ”€â”€ 015_docker_cleanup.go
â”‚       â”œâ”€â”€ 016_waf.go
â”‚       â”œâ”€â”€ 017_waf_enhance.go
â”‚       â”œâ”€â”€ 018_backup_uploads.go
â”‚       â”œâ”€â”€ 019_cron_storage.go
â”‚       â””â”€â”€ 020_incremental_backups.go
â”‚
â”œâ”€â”€ config/                    # Configuration
â”‚   â””â”€â”€ config.go             # JSON config reader/writer
â”‚
â”œâ”€â”€ pkg/                      # Shared utilities
â”‚   â””â”€â”€ ...                   # Crypto, logger, errors
â”‚
â”œâ”€â”€ i18n/                     # Backend i18n
â”‚   â””â”€â”€ ...                   # EN/VI translations
â”‚
â”œâ”€â”€ task/                     # Background task queue
â”‚   â””â”€â”€ queue.go             # Async task with progress
â”‚
â”œâ”€â”€ websocket/               # WebSocket hub
â”‚   â””â”€â”€ ...                  # Connection management
â”‚
â”œâ”€â”€ system/                   # System utilities
â”‚   â”œâ”€â”€ shell.go            # Shell execution
â”‚   â””â”€â”€ os_detect.go       # OS detection
â”‚
â””â”€â”€ plugin/                  # Plugin system
```

```
    â”œâ”€â”€ plugin.go          # Plugin interface
    â””â”€â”€ manager.go         # Plugin loader

cmd/
â””â”€â”€ panel/
    â””â”€â”€ main.go            # Entry point
```

Slide 5: Authentication & Authorization

JWT + TOTP 2FA + RBAC

Authentication Flow

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”,                       Authentication Pipeline                                â”,
â”,                                                                              â”,
â”,  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”      â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”      â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”      â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”       â”,
â”,  â”,  Login â”, â”€â”€â†’  â”,  Verify â”, â”€â”€â†’  â”, Generateâ”, â”€â”€â†’  â”, Return  â”,       â”,
â”,  â”, Requestâ”,      â”,Password â”,      â”,  Token  â”,      â”,  Tokens â”,       â”,
â”,  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜      â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜      â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜      â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜       â”,
â”,       â”,                  â”,                â”,                                  â”,
â”,       â”,              bcrypt            JWT HS256                          â”,
â”,       â”,              cost=12        â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”                        â”,
â”,       â”,                       â”Œâ”€â”€â”€â”€â”, Access Token â”,                   â”,
â”,       â”,                       â”,    â”, (15 min)     â”,                      â”,
â”,       â”,                       â”,    â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜                        â”,
â”,       â”,                       â”,    â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”                        â”,
â”,       â”,                       â””â”€â”€â”€â”€â”,Refresh Tokenâ”,                   â”,
â”,       â”,                            â”, (7 days)    â”,                      â”,
â”,       â”,                            â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜                       â”,
â”,       â–¼                                                                â”,
â”,  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”        â”,
â”,  â”,  2FA Check (if enabled)                                 â”,         â”,
â”,  â”,  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”      â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”      â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”,        â”,
â”,  â”,  â”,  TOTP   â”, â”€â”€â†’  â”,   Validate  â”, â”€â”€â†’  â”, Update Login â”,   â”,        â”,
â”,  â”,  â”,  Code   â”,      â”, 6-digit RFC â”,      â”,   Status     â”,   â”,        â”,
â”,  â”,  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜      â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜      â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”,        â”,
â”,  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜        â”,
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

JWT Token Structure

```
// Access Token Claims (15 minutes TTL)
type AccessClaims struct {
    UserID      uint     `json:"user_id"`
    Username    string   `json:"username"`
    Email       string   `json:"email"`
    Role        string   `json:"role"`
    Permissions []string `json:"permissions"`  // Cached permissions
    TokenType   string   `json:"type"`         // "access"
    RegisteredClaims
}

// Refresh Token Claims (7 days TTL)
type RefreshClaims struct {
    UserID    uint   `json:"user_id"`
    TokenType string `json:"type"`  // "refresh"
    jti       string `json:"jti"`  // Unique token ID for revocation
    RegisteredClaims
}
```

RBAC Permission Model

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”,  Permission Namespace System                                    â”,
â”,                                                                 â”,
â”,  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”,
â”,  â”,  Admin Role: ["*"]                                    â”,   â”,
â”,  â”,  â”œâ”€â”€ Full access to all resources                         â”,   â”,
â”‚  â”‚  â””â”€â”€ Can manage users and roles                          â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â”‚                                                                 â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”‚  Operator Role:                                           â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ sites.*           (all site operations)            â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ databases.*       (all database operations)        â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ files.*           (all file operations)           â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ docker.*         (all docker operations)          â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ cron.*            (all cron operations)           â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ backup.*         (all backup operations)          â”‚   â”‚
â”‚  â”‚  â””â”€â”€ terminal.access  (terminal access)               â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â”‚                                                                 â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”‚  Viewer Role:                                            â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ sites.read         (read-only site view)           â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ databases.read     (read-only database view)       â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ files.read         (read-only file view)          â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ logs.read          (read-only log view)           â”‚   â”‚
â”‚  â”‚  â””â”€â”€ docker.read        (read-only docker view)         â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

TOTP 2FA Implementation

```
// TOTP Configuration (RFC 6238)
const (
    TOTPIssuer      = "ITDLUPanel"
    TOTPDigits      = 6
    TOTPPeriod      = 30  // seconds
    TOTPSecretSize  = 20  // bytes
)

// Enable 2FA - Generate QR Code Secret
func (s *AuthService) EnableTOTP(userID uint) (*TOTPResponse, error) {
    // Generate random secret
    secret, err := totp.Generate(totp.GenerateOpts{
        Issuer:      TOTPIssuer,
        AccountName: username,
        SecretSize:  TOTPSecretSize,
        Period:      TOTPPeriod,
        Digits:      otp.DigitsSix,
        Algorithm:   otp.AlgorithmSHA1,
    })

    // Generate provisioning URI for QR code
    provisioningURI := secret.URL()

    return &TOTPResponse{
        Secret:     secret.Secret(),
        ProvisioningURI: provisioningURI,
        QRCode:     generateQRCodeBase64(provisioningURI),
    }
}

// Verify TOTP Code
func (s *AuthService) VerifyTOTP(secret, code string) bool {
    return totp.Validate(code, secret)
}
```

Slide 6: Database Schema Chi Tiáº¿t

SQLite Database Architecture

Entity Relationship Diagram

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                         Core User & Auth Entities                            â”‚
â”‚                                                                             â”‚
```

```
â”‚    â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”         â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”         â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”           â”‚
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```

Key Table Definitions

Users Table

```
CREATE TABLE users (
    id              INTEGER PRIMARY KEY AUTOINCREMENT,
    username        TEXT UNIQUE NOT NULL,
    email           TEXT UNIQUE,
    password_hash   TEXT NOT NULL,          -- bcrypt cost=12
    totp_secret     TEXT,                    -- base32 encoded
    totp_enabled    INTEGER DEFAULT 0,
    status          INTEGER DEFAULT 1,       -- 1=active, 0=disabled
    last_login      DATETIME,
    created_at      DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at      DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Indexes
CREATE INDEX idx_users_username ON users(username);
CREATE INDEX idx_users_email ON users(email);
```

Roles Table

```
CREATE TABLE roles (
    id          INTEGER PRIMARY KEY AUTOINCREMENT,
    name        TEXT UNIQUE NOT NULL,
    description TEXT,
    permissions TEXT NOT NULL,              -- JSON array ["sites.*", "docker.*"]
    is_system   INTEGER DEFAULT 0,          -- 1=system role (cannot delete)
    created_at  DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- System roles
INSERT INTO roles (name, permissions, is_system) VALUES
    ('admin', '["*"]', 1),
    ('operator', '["sites.*","databases.*","files.*","docker.*","cron.*","backup.*","terminal.access"]', 1),
    ('viewer', '["sites.read","databases.read","files.read","logs.read","docker.read"]', 1);
```

Audit Logs Table

```
CREATE TABLE audit_logs (
    id          INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id     INTEGER REFERENCES users(id),
    username    TEXT,
    action      TEXT NOT NULL,               -- CREATE, UPDATE, DELETE, LOGIN, etc.
    resource    TEXT,                        -- sites, databases, users, etc.
    details     TEXT,                        -- JSON with additional details
    ip_address  TEXT,
    created_at  DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Index for efficient querying
CREATE INDEX idx_audit_user ON audit_logs(user_id);
CREATE INDEX idx_audit_created ON audit_logs(created_at);
CREATE INDEX idx_audit_action ON audit_logs(action);
```

Slide 7: Docker Integration Chi Tiáº¿t

Docker Management System

Architecture Overview

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â”‚   â”‚  â”‚ â€¢ Exec/Attach   â”‚          â”‚  Prometheus, Grafana...    â”‚  â”‚      â”‚
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â”‚   â”‚  â”‚              Docker API Client                       â”‚    â”‚      â”‚
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```

Docker API Endpoints

Container Management (20+ endpoints)

```
// Container CRUD
GET    /api/v1/docker/containers              // List all containers
POST   /api/v1/docker/containers              // Create container
GET    /api/v1/docker/containers/:id         // Inspect container
DELETE /api/v1/docker/containers/:id         // Remove container

// Container Lifecycle
POST   /api/v1/docker/containers/:id/start     // Start container
POST   /api/v1/docker/containers/:id/stop      // Stop container
POST   /api/v1/docker/containers/:id/restart    // Restart container
POST   /api/v1/docker/containers/:id/pause      // Pause container
POST   /api/v1/docker/containers/:id/unpause    // Unpause container

// Container Operations
GET    /api/v1/docker/containers/:id/logs    // Get container logs
GET    /api/v1/docker/containers/:id/stats   // Get resource stats
POST   /api/v1/docker/containers/:id/rename  // Rename container
POST   /api/v1/docker/containers/:id/exec    // Execute command
POST   /api/v1/docker/containers/:id/commit  // Create image from container
```

Image Management (6 endpoints)

```
GET    /api/v1/docker/images                // List all images
POST   /api/v1/docker/images/pull           // Pull image from registry
DELETE /api/v1/docker/images/:id           // Remove image
GET    /api/v1/docker/images/:id           // Inspect image
GET    /api/v1/docker/images/search        // Search Docker Hub
POST   /api/v1/docker/images/:id/tag       // Tag image
```

Network Management (7 endpoints)

```
GET    /api/v1/docker/networks              // List networks
POST   /api/v1/docker/networks            // Create network
DELETE /api/v1/docker/networks/:id         // Remove network
GET    /api/v1/docker/networks/:id         // Inspect network
POST   /api/v1/docker/networks/:id/connect   // Connect container
POST   /api/v1/docker/networks/:id/disconnect // Disconnect container
```

Volume Management (5 endpoints)

```
GET    /api/v1/docker/volumes              // List volumes
POST   /api/v1/docker/volumes             // Create volume
DELETE /api/v1/docker/volumes/:name       // Remove volume
GET    /api/v1/docker/volumes/:name       // Inspect volume
POST   /api/v1/docker/volumes/prune       // Prune unused volumes
```

Docker Compose (5 endpoints)

```
GET    /api/v1/docker/compose              // List compose projects
POST   /api/v1/docker/compose             // Create compose project
POST   /api/v1/docker/compose/:name/up     // Start project
POST   /api/v1/docker/compose/:name/down  // Stop project
DELETE /api/v1/docker/compose/:name       // Remove project
```

System (5 endpoints)

```
GET    /api/v1/docker/info                // Docker engine info
GET    /api/v1/docker/daemon/config      // Get daemon.json
POST   /api/v1/docker/daemon/config      // Save daemon.json
POST   /api/v1/docker/restart            // Restart Docker
POST   /api/v1/docker/stop               // Stop Docker
```

OneClick App Templates

```
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â”‚  â”‚ â€¢ Alertmanager            â”‚ â€¢ Loki (Log aggregation)                  â”‚  â”‚
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â”‚  â”‚ â€¢ Nextcloud (File Sync)   â”‚ â€¢ Mattermost                              â”‚  â”‚
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â”‚  â”‚ â€¢ Emby (Media)           â”‚ â€¢ Invoice Ninja                        â”‚  â”‚
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â”‚  â”‚ â€¢ pgAdmin                â”‚                                        â”‚  â”‚
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Slide 8: WebSocket Real-Time Communication

WebSocket Architecture

Dual WebSocket Endpoints

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â”‚                         WebSocket Connection Flow                          â”‚
â”‚                                                                            â”‚
â”‚   Browser (xterm.js / React)                                               â”‚
â”‚       â”‚                                                                    â”‚
â”‚       â”‚  1. Connect to /api/v1/terminal                                    â”‚
â”‚       â”‚  2. Send: { "type": "resize", "cols": 80, "rows": 24 }             â”‚
â”‚       â”‚  3. Send: { "type": "input", "data": "ls -la\r" }                  â”‚
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```

Terminal WebSocket Protocol

```
// Client â†’ Server Messages
type TerminalInput struct {
    Type string `json:"type"`  // "input"
    Data string `json:"data"`  // Keystrokes
}

type TerminalResize struct {
    Type  string `json:"type"`  // "resize"
    Cols  int    `json:"cols"`
    Rows  int    `json:"rows"`
}
```

```
// Server â†’ Client Messages
type TerminalOutput struct {
    Type string `json:"type"`  // "output"
    Data string `json:"data"`  // Terminal output (base64 or raw)
}

// Backend Implementation
func (h *TerminalHub) HandleWebSocket(c *gin.Context) {
    // Upgrade HTTP to WebSocket
    conn, err := upgrader.Upgrade(c.Writer, c.Request, nil)
    defer conn.Close()

    // Create PTY
    f, tty, err := pty.StartWithSize(cmd, nil)
    defer f.Close()

    // Bidirectional copy
    go func() {
        io.Copy(f, conn)  // Client â†’ PTY
    }()
    io.Copy(conn, f)     // PTY â†’ Client
}
```

Monitor WebSocket Protocol

```
// Server â†’ Client (every 2 seconds)
type MetricsUpdate struct {
    Type     string    `json:"type"`      // "metrics"
    Timestamp time.Time `json:"timestamp"`
    CPU       CPUMetrics  `json:"cpu"`
    Memory    MemMetrics  `json:"memory"`
    Disk      DiskMetrics `json:"disk"`
    Network   NetMetrics  `json:"network"`
}

type CPUMetrics struct {
    Percent   float64   `json:"percent"`   // 0-100
    User      float64   `json:"user"`
    System    float64   `json:"system"`
    Idle      float64   `json:"idle"`
    CoreCount int       `json:"core_count"`
}

type MemMetrics struct {
    Total       uint64  `json:"total"`      // bytes
    Used        uint64  `json:"used"`
    Available   uint64  `json:"available"`
    UsedPercent float64 `json:"used_percent"`
}

type DiskMetrics struct {
    Total       uint64  `json:"total"`
    Used        uint64  `json:"used"`
    Free        uint64  `json:"free"`
    UsedPercent float64 `json:"used_percent"`
}

type NetMetrics struct {
    BytesSent   uint64 `json:"bytes_sent"`
    BytesRecv   uint64 `json:"bytes_recv"`
    PacketsSent uint64 `json:"packets_sent"`
    PacketsRecv uint64 `json:"packets_recv"`
}
```

Slide 9: File Management System

File Manager Architecture

Capabilities Overview

```
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```

Monaco Editor Integration

```
// Monaco Editor Supported Languages (50+)
var supportedLanguages = []string{
    // Web
    "javascript", "typescript", "html", "css", "scss", "less",
    "json", "xml", "yaml", "markdown",

    // Backend
    "go", "python", "java", "c", "cpp", "csharp", "ruby", "php", "rust",

    // Config
    "ini", "properties", "dotenv", "nginx", "apache",

    // Shell
    "shell", "powershell", "dockerfile",

    // Database
    "sql", "plsql", "mysql",

    // Data
    "csv", "toml",
}
```

File API Endpoints

```
// Navigation & Info
GET  /api/v1/files                  // List directory contents
GET  /api/v1/files/info             // Get file/directory info
GET  /api/v1/files/tree            // Get directory tree

// Read & Write
GET  /api/v1/files/read            // Read file content
POST /api/v1/files/save            // Save file content

// Create & Delete
POST /api/v1/files/create           // Create file or directory
DELETE /api/v1/files                // Delete files/directories

// Modify
```

```
POST /api/v1/files/rename          // Rename file/directory
POST /api/v1/files/copy            // Copy file/directory
POST /api/v1/files/move            // Move file/directory
POST /api/v1/files/permissions     // Change permissions (chmod)

// Transfer
POST /api/v1/files/upload          // Upload file (multipart)
GET  /api/v1/files/download        // Download file
POST /api/v1/files/remote-download // Download from URL

// Archive
POST /api/v1/files/compress        // Compress to ZIP/TAR/TARGZ
POST /api/v1/files/decompress      // Extract archive

// Search
POST /api/v1/files/search          // Search files
```

Slide 10: Backup System Architecture

Multi-Destination Backup System

Backup Flow

```
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â”,   â”,   Request    â”,                                                          â”,
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â”,   â”,                      Backup Service                               â”,     â”,
â”,   â”,                                                                   â”,     â”,
â”,   â”,  1. Determine backup type (full/incremental)                      â”,     â”,
â”,   â”,  2. Create temporary directory                                    â”,     â”,
â”,   â”,  3. Execute backup (mysqldump, tar, etc.)                         â”,     â”,
â”,   â”,  4. Calculate checksum (SHA256)                                   â”,     â”,
â”,   â”,  5. Encrypt if enabled (AES-256-GCM)                              â”,     â”,
â”,   â”,  6. Compress (gzip)                                               â”,     â”,
â”,   â”,  7. Upload to destination(s)                                      â”,     â”,
â”,   â”,  8. Record metadata in database                                   â”,     â”,
â”,   â”,  9. Cleanup old backups                                           â”,     â”,
â”,   â”,                                                                   â”,     â”,
â”,   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜     â”,
â”,                                    â”,                                      â”,
â”,          â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¼â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”            â”,
â”,          â”,                           â”,                           â”,            â”,
â”,          â–¼                           â–¼                           â–¼            â”,
â”,   â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”          â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”          â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”     â”,
â”,   â”,    Local     â”,          â”,   Network    â”,          â”,    Cloud     â”,     â”,
â”,   â”,   Storage    â”,          â”,   Storage    â”,          â”,   Storage    â”,     â”,
â”,   â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤          â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤          â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤     â”,
â”,   â”, /ITDLUPanel â”,          â”, FTP/SFTP     â”,          â”, Amazon S3    â”,     â”,
â”,   â”, /backups    â”,          â”,              â”,          â”, MinIO        â”,     â”,
â”,   â”,             â”,          â”,              â”,          â”,              â”,     â”,
â”,   â”,             â”,          â”,              â”,          â”, Google Drive â”,     â”,
â”,   â”,             â”,          â”,              â”,          â”,              â”,     â”,
â”,   â”,             â”,          â”,              â”,          â”, OneDrive     â”,     â”,
â”,   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜          â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜          â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜     â”,
â”,                                                                             â”,
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Backup Types

Full Backup

```
# Database (MySQL)
mysqldump --single-transaction --routines --triggers \
    --events --all-databases | gzip > backup_full.sql.gz

# Files
tar -czf backup_files.tar.gz /var/www/sites/
```

Incremental Backup

```
// Incremental backup using rsync-style diff
type IncrementalBackup struct {
    BaseBackupID  uint      `db:"base_backup_id"`
    DiffPath      string    `db:"diff_path"`
    ChangesCount  int       `db:"changes_count"`
    ChangesSize   int64     `db:"changes_size"`
    CreatedAt     time.Time `db:"created_at"`
}

// Only backup changed files since last backup
func (s *BackupService) CreateIncremental(ctx context.Context, siteID uint) error {
    lastBackup := s.getLastBackup(siteID)
    lastBackupTime := lastBackup.CreatedAt

    // Find all modified files
    changes, err := s.findChanges(siteID, lastBackupTime)

    // Create diff archive
    for _, change := range changes {
        s.addToDiff(change)
    }

    return nil
}
```

Encryption

```
// AES-256-GCM Encryption
type EncryptedBackup struct {
    Ciphertext   []byte
    Nonce        []byte    // 12 bytes
    Tag          []byte    // 16 bytes
}

// Encrypt backup
func EncryptBackup(data []byte, password string) (*EncryptedBackup, error) {
    key := deriveKey(password)  // PBKDF2 with 100,000 iterations

    block, err := aes.NewCipher(key)
    gcm, err := cipher.NewGCM(block)

    nonce := make([]byte, gcm.NonceSize())
    io.ReadFull(rand.Reader, nonce)

    sealed := gcm.Seal(nonce, nonce, data, nil)

    return &EncryptedBackup{
        Ciphertext: sealed[len(nonce):],
        Nonce:      nonce,
        Tag:        sealed[len(nonce):],
    }, nil
}
```

Slide 11: SSL Certificate Management

Let's Encrypt Integration

Certificate Lifecycle

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                      SSL Certificate Lifecycle                              â”‚
â”‚                                                                             â”‚
â”‚   â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”     â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”     â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”     â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”             â”‚
```

```
â”‚   â”‚ Request â”‚â”€â”€â”€â”€â–¶â”‚  Issue  â”‚â”€â”€â”€â”€â–¶â”‚ Install â”‚â”€â”€â”€â”€â–¶â”‚  Renew  â”‚            â”‚
â”‚   â”‚  Cert   â”‚     â”‚  Cert   â”‚     â”‚  Cert   â”‚     â”‚  (auto) â”‚            â”‚
â”‚   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜     â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜     â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜     â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜            â”‚
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â”‚        â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚              â”‚
â”‚        â”‚  â”‚  ACME Protocol Flow                          â”‚   â”‚              â”‚
â”‚        â”‚  â”‚                                              â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  1. GET /directory (ACME server)            â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  2. POST /new-account                       â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  3. POST /new-order (domain validation)     â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  4. ACME Challenge (http-01 or dns-01)      â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  5. POST /challenge (validation)            â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  6. POST /chalange (poll status)            â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  7. GET /order/:id (certificate ready)      â”‚   â”‚              â”‚
â”‚        â”‚  â”‚  8. POST /order/:id/cert (download cert)    â”‚   â”‚              â”‚
â”‚        â”‚  â”‚                                              â”‚   â”‚              â”‚
â”‚        â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚              â”‚
â”‚        â”‚                                                     â”‚             â”‚
â”‚        â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜             â”‚
â”‚                                                                              â”‚
â”‚   â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”  â”‚
â”‚   â”‚  Auto-Renewal Schedule                                           â”‚  â”‚
â”‚   â”‚                                                                    â”‚  â”‚
â”‚   â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”    â”‚  â”‚
â”‚   â”‚  â”‚  Certificate Lifetime: 90 days                            â”‚    â”‚  â”‚
â”‚   â”‚  â”‚  Auto-Renew: 30 days before expiry                        â”‚    â”‚  â”‚
â”‚   â”‚  â”‚  Check Schedule: Daily at 00:00 UTC                       â”‚    â”‚  â”‚
â”‚   â”‚  â”‚  Renewal Process: Silent (no user interaction required)    â”‚    â”‚  â”‚
â”‚   â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜    â”‚  â”‚
â”‚   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â”‚
â”‚                                                                              â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Certificate Storage

```
// Certificate Storage Structure
type SSLCertificate struct {
    Domain         string    `json:"domain"`
    CertPath       string    `json:"cert_path"`       // /etc/ssl/certs/domain.crt
    KeyPath        string    `json:"key_path"`        // /etc/ssl/private/domain.key
    ChainPath      string    `json:"chain_path"`      // /etc/ssl/certs/domain.chain.crt
    Provider       string    `json:"provider"`        // "letsencrypt"
    ValidFrom      time.Time `json:"valid_from"`
    ValidUntil     time.Time `json:"valid_until"`
    AutoRenew      bool      `json:"auto_renew"`
    LastRenewed    time.Time `json:"last_renewed"`
}

// Nginx Configuration
const nginxSSLConfig = `
server {
    listen 443 ssl http2;
    server_name {{.Domain}};

    ssl_certificate     {{.CertPath}};
    ssl_certificate_key {{.KeyPath}};
    ssl_trusted_certificate {{.ChainPath}};

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256;
    ssl_prefer_server_ciphers on;
    ssl_session_cache shared:SSL:10m;

    # Security headers
    add_header Strict-Transport-Security "max-age=31536000" always;
    add_header X-Frame-Options DENY;
    add_header X-Content-Type-Options nosniff;
}
`
```

Slide 12: Frontend Architecture

React 19 SPA Architecture

Component Hierarchy

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                         Frontend Architecture                                â”‚
â”‚                                                                              â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”‚
â”‚  â”‚  App.tsx (Root)                                                         â”‚ â”‚
â”‚  â”‚  â”œâ”€â”€ AuthProvider (Context)                                            â”‚ â”‚
â”‚  â”‚  â”‚   â””â”€â”€ Login Page (unauthenticated routes)                           â”‚ â”‚
â”‚  â”‚  â”‚                                                                      â”‚ â”‚
â”‚  â”‚  â””â”€â”€ MainLayout                                                        â”‚ â”‚
â”‚  â”‚      â”œâ”€â”€ Sidebar (Navigation)                                          â”‚ â”‚
â”‚  â”‚      â”œâ”€â”€ Header (User menu, notifications)                             â”‚ â”‚
â”‚  â”‚      â””â”€â”€ Content Area                                                  â”‚ â”‚
â”‚  â”‚          â”‚                                                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ Lazy-loaded Pages:                                        â”‚ â”‚
â”‚  â”‚          â”‚                                                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /dashboard     â†’ DashboardPage                            â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /sites         â†’ WebsitesPage                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /databases    â†’ DatabasesPage                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /files         â†’ FilesPage (15 components)                â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /terminal      â†’ TerminalPage                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /docker        â†’ DockerPage (10 components)               â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /security      â†’ SecurityPage                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /firewall      â†’ FirewallPage                             â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /monitor       â†’ MonitorPage (3 charts)                   â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /cron          â†’ CronPage                                 â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /logs           â†’ LogsPage                                â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /appstore       â†’ AppStorePage                            â”‚ â”‚
â”‚  â”‚          â”œâ”€â”€ /storage        â†’ StoragePage                             â”‚ â”‚
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```

Key React Patterns

State Management

```
// Zustand Store Example
interface AuthState {
  user: User | null;
  accessToken: string | null;
  refreshToken: string | null;
  isAuthenticated: boolean;
  // Actions
  login: (credentials: LoginCredentials) => Promise<void>;
  logout: () => void;
  refreshTokens: () => Promise<void>;
  updateProfile: (data: Partial<User>) => void;
}

export const useAuthStore = create<AuthState>((set, get) => ({
  user: null,
  accessToken: null,
  refreshToken: null,
  isAuthenticated: false,

  login: async (credentials) => {
    const response = await api.post('/auth/login', credentials);
    const { user, access_token, refresh_token } = response.data;

    set({
      user,
      accessToken: access_token,
      refreshToken: refresh_token,
      isAuthenticated: true
    });

    // Store in localStorage for persistence
    localStorage.setItem('refresh_token', refresh_token);
  },

  logout: () => {
    api.post('/auth/logout');
    set({ user: null, accessToken: null, refreshToken: null, isAuthenticated: false });
    localStorage.removeItem('refresh_token');
  },
}));
```

Theme System

```
// usePalette hook
export function usePalette() {
  const isDark = document.body.getAttribute('data-theme') === 'dark';

  return {
    isDark,
    p: {
      bg: isDark ? '#0D1117' : '#F5F7FA',
      card: isDark ? '#161B22' : '#FFFFFF',
      cardBorder: isDark ? 'rgba(255,255,255,0.1)' : '#E8EDF2',
      text: isDark ? '#E6EDF3' : '#1A1A2E',
      textMuted: isDark ? '#8B949E' : '#64748B',
      headerBg: isDark ? '#21262D' : '#FAFBFC',
      accent: '#1565C0',
      green: '#52C41A',
      red: '#FF4D4F',
      amber: '#FAAD14',
      purple: '#722ED1',
      cyan: '#13C2C2',
    }
  };
}
```

Slide 13: Performance Optimization

Performance & Optimization

Backend Optimizations

Database Connection Pooling

```
// SQLite connection pool settings
type DBConfig struct {
    MaxOpenConns    int = 25        // Maximum open connections
    MaxIdleConns    int = 5         // Maximum idle connections
    ConnMaxLifetime time.Duration = time.Hour
}

// Prepared statements for frequently used queries
type PreparedStatements struct {
    GetUserByUsername *sql.Stmt
    GetSitesByUser    *sql.Stmt
    GetContainers     *sql.Stmt
    InsertAuditLog    *sql.Stmt
}
```

Concurrent Request Handling

```
// Gin context pool for memory optimization
engine.Use(gin.Logger())
engine.Use(gin.Recovery())

// Custom context pool
engine.Use(func(c *gin.Context) {
    c.Next()
})

// HTTP server settings
s := &http.Server{
    Addr:         ":8888",
    Handler:      engine,
    ReadTimeout:  30 * time.Second,
    WriteTimeout: 30 * time.Second,
    IdleTimeout:  120 * time.Second,
    MaxHeaderBytes: 1 << 20, // 1MB
}
```

Caching Strategy

```
// Permission cache with TTL
type PermissionCache struct {
    mu    sync.RWMutex
    data  map[uint][]string
    timer map[uint]*time.Timer
    ttl   time.Duration
}

func (c *PermissionCache) Get(userID uint) ([]string, bool) {
    c.mu.RLock()
    defer c.mu.RUnlock()
    perms, ok := c.data[userID]
    return perms, ok
}

func (c *PermissionCache) Set(userID uint, perms []string) {
    c.mu.Lock()
    defer c.mu.Unlock()

    c.data[userID] = perms

    // Reset timer
    if c.timer[userID] != nil {
        c.timer[userID].Stop()
    }
    c.timer[userID] = time.AfterFunc(c.ttl, func() {
        c.Delete(userID)
    })
}
```

Frontend Optimizations

Code Splitting

```
// React.lazy for all pages
const Dashboard = lazy(() => import('./pages/Dashboard'));
const Websites = lazy(() => import('./pages/Websites'));
const Docker = lazy(() => import('./pages/Docker'));
const Files = lazy(() => import('./pages/Files'));

// Route-level splitting
<Suspense fallback={<Spin />}>
  <Routes>
    <Route path="/dashboard" element={<Dashboard />} />
    <Route path="/docker/*" element={<Docker />} />
  </Routes>
</Suspense>
```

Memoization

```
// Heavy component memoization
const HeavyTable = memo(({ data, columns }: TableProps) => {
  return (
    <Table
      dataSource={data}
      columns={useMemo(() => columns, [columns])}
      pagination={false}
    />
  );
});

// Expensive computation memoization
const processedData = useMemo(() => {
  return data.map(item => transformItem(item));
}, [data]);
```

WebSocket Optimization

```
// Debounced metrics updates
const MetricsUpdater = () => {
  const [metrics, setMetrics] = useState<Metrics>(null);

  useEffect(() => {
    let buffer: Metrics[] = [];
    let rafId: number;

    const processBuffer = () => {
      // Take latest from buffer, ignore others
      if (buffer.length > 0) {
        setMetrics(buffer[buffer.length - 1]);
        buffer = [];
      }
      rafId = requestAnimationFrame(processBuffer);
    };

    ws.onmessage = (event) => {
      buffer.push(JSON.parse(event.data));
    };

    rafId = requestAnimationFrame(processBuffer);
    return () => cancelAnimationFrame(rafId);
  }, []);
};
```

Slide 14: Security Implementation

Security Hardening

Security Layers

```
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â”‚   â”‚  â”œâ”€â”€ TLS 1.2/1.3 only                                                â”‚ â”‚
â”‚   â”‚  â”œâ”€â”€ Strong cipher suites                                             â”‚ â”‚
```

```
â”‚   â”‚  â”œâ”€â”€ HSTS (HTTP Strict Transport Security)                           â”‚ â”‚
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```

Security Headers

```
// Middleware: SecurityHeaders
func SecurityHeadersMiddleware() gin.HandlerFunc {
    return func(c *gin.Context) {
        // Prevent clickjacking
        c.Header("X-Frame-Options", "DENY")

        // Prevent MIME sniffing
        c.Header("X-Content-Type-Options", "nosniff")

        // XSS filter (legacy but still useful)
        c.Header("X-XSS-Protection", "1; mode=block")

        // Content Security Policy
        c.Header("Content-Security-Policy",
            "default-src 'self'; script-src 'self' 'unsafe-inline'; style-src 'self' 'unsafe-inline';")

        // Referrer Policy
        c.Header("Referrer-Policy", "strict-origin-when-cross-origin")

        c.Next()
    }
}
```

Password Security

```
// bcrypt with cost=12
const bcryptCost = 12

func HashPassword(password string) (string, error) {
    bytes, err := bcrypt.GenerateFromPassword([]byte(password), bcryptCost)
    return string(bytes), err
}

func CheckPassword(password, hash string) bool {
    err := bcrypt.CompareHashAndPassword([]byte(hash), []byte(password))
    return err == nil
}

// Password requirements
const passwordMinLength = 8
const passwordRequireUppercase = true
const passwordRequireLowercase = true
const passwordRequireNumber = true
const passwordRequireSpecial = true
```

Rate Limiting

```
// Token bucket rate limiter
type RateLimiter struct {
    tokens     float64
    maxTokens  float64
    refillRate float64  // tokens per second
    lastRefill time.Time
    mu         sync.Mutex
}

func (rl *RateLimiter) Allow() bool {
    rl.mu.Lock()
    defer rl.mu.Unlock()

    // Refill tokens
    now := time.Now()
    elapsed := now.Sub(rl.lastRefill).Seconds()
    rl.tokens = min(rl.maxTokens, rl.tokens+elapsed*rl.refillRate)
    rl.lastRefill = now

    // Check if request allowed
    if rl.tokens >= 1 {
        rl.tokens--
        return true
    }
    return false
}

// Configuration: 300 requests per minute = 5 requests per second
rateLimiter := &RateLimiter{
    maxTokens:  300,
    refillRate: 5,
}
```

Slide 15: CI/CD & Deployment

Build & Deployment Pipeline

Build Process

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                          Build Pipeline                                      â”‚
â”‚                                                                             â”‚
â”‚   â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”‚
â”‚   â”‚  Phase 1: Frontend Build (Node.js 22)                                  â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
â”‚   â”‚  $ cd web && npm install                                              â”‚ â”‚
â”‚   â”‚  $ npx vite build                                                     â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
â”‚   â”‚  Output: web/dist/                                                    â”‚ â”‚
â”‚   â”‚  â”œâ”€â”€ index.html                                                       â”‚ â”‚
â”‚   â”‚  â”œâ”€â”€ assets/                                                          â”‚ â”‚
â”‚   â”‚  â””â”€â”€ favicon.ico                                                      â”‚ â”‚
â”‚   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”‚
â”‚                                    â”‚                                         â”‚
â”‚                                    â–¼                                         â”‚
â”‚   â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”‚
â”‚   â”‚  Phase 2: Go Embed & Compile                                          â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
â”‚   â”‚  // main.go                                                           â”‚ â”‚
â”‚   â”‚  //go:embed web/dist                                                 â”‚ â”‚
â”‚   â”‚  var staticFS embed.FS                                               â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
â”‚   â”‚  $ CGO_ENABLED=0 GOOS=linux GOARCH=amd64 \                           â”‚ â”‚
â”‚   â”‚      go build -ldflags="-s -w" -o bin/itdlupanel ./cmd/panel         â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
â”‚   â”‚  Output: bin/itdlupanel (~60MB, static binary)                       â”‚ â”‚
â”‚   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”‚
â”‚                                                                             â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Deployment Options

1. Direct Binary

```
# Download and install
wget https://github.com/itdlu/ITDLUPanel/releases/latest/itdlupanel
chmod +x itdlupanel
sudo mv itdlupanel /usr/local/bin/

# Create systemd service
sudo tee /etc/systemd/system/itdlupanel.service <<EOF
[Unit]
Description=ITDLUPanel Server
After=network.target

[Service]
Type=simple
User=root
WorkingDirectory=/ITDLUPanel
ExecStart=/usr/local/bin/itdlupanel -config /ITDLUPanel/config/panel.json
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
EOF

# Enable and start
sudo systemctl daemon-reload
sudo systemctl enable itdlupanel
sudo systemctl start itdlupanel
```

2. Docker Deployment

```
# Dockerfile
FROM alpine:3.21

RUN apk add --no-cache ca-certificates tzdata

COPY itdlupanel /usr/local/bin/
RUN chmod +x /usr/local/bin/itdlupanel

EXPOSE 8888

USER 1000

ENTRYPOINT ["itdlupanel"]
CMD ["-config", "/data/panel.json"]

VOLUME ["/data"]
```

```
# Run with Docker
docker run -d \
  --name itdlupanel \
  -p 8888:8888 \
  -v /ITDLUPanel/data:/data \
  -v /var/run/docker.sock:/var/run/docker.sock \
  --restart unless-stopped \
  itdlu/itdlupanel:latest
```

Slide 16: Code Statistics

Project Metrics

Backend (Go)

Metric	Count
Total Go Files	112
Total Lines of Code	43,314

API Handlers	24 files
Services	33+
Installers	14
Vhost Generators	4
Database Migrations	14

LOC Breakdown by Layer

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚  Backend Code Distribution                                                  â”‚
â”‚                                                                             â”‚
â”‚  Service Layer          â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ  23,437 LOC   â”‚
â”‚  API Handlers          â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ  9,814 LOC                            â”‚
â”‚  Database/Migrations   â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ  5,234 LOC                                â”‚
â”‚  Installers            â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ  4,123 LOC                                  â”‚
â”‚  Other (config, i18n)  â–ˆâ–ˆâ–ˆâ–ˆ  2,706 LOC                                   â”‚
â”‚                                                                             â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Frontend (React/TypeScript)

Metric	Count
Total TSX/TS Files	93
Pages	17 (lazy-loaded)
Shared Components	15
Zustand Stores	5
Custom Hooks	7+

Frontend Code Distribution

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚  Frontend Code Distribution                                                  â”‚
â”‚                                                                             â”‚
â”‚  Pages                  â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ          â”‚
â”‚  Components             â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ                           â”‚
â”‚  Hooks/Utils            â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ                                        â”‚
â”‚  Stores                 â–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆâ–ˆ                                             â”‚
â”‚  Types                  â–ˆâ–ˆâ–ˆâ–ˆâ–ˆ                                                â”‚
â”‚                                                                             â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

API Coverage

Module	REST Endpoints	WebSocket
Auth	9	-
Dashboard	2	-
Sites	14	-
Databases	8	-
Docker	46+	-
Files	17	-
Security	12	-
Cron	8	-
Logs	7	-
SSL	9	-
Backup	6	-
Monitor	5	1
Terminal	-	1
Total	**~160**	**2**

Slide 17: Testing Strategy

Quality Assurance

Testing Pyramid

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                              Testing Pyramid                                 â”‚
â”‚                                                                             â”‚
â”‚                                    â–²                                         â”‚
â”‚                                   â•± â•²                                        â”‚
â”‚                                  â•±   â•²                                       â”‚
â”‚                                 â•±     â•²                                      â”‚
â”‚                                â•± Test  â•²                                     â”‚
â”‚                               â•± Pyramid â•²                                    â”‚
â”‚                              â•±           â•²                                   â”‚
â”‚                             â•±â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â•²                                  â”‚
â”‚                            â•±               â•²                                 â”‚
â”‚                           â•±â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â•²                                â”‚
â”‚                                                                             â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”‚  E2E Tests (few, high confidence)                                    â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ Selenium/Playwright                                              â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ Full user flows                                                  â”‚   â”‚
â”‚  â”‚  â””â”€â”€ Cross-browser testing                                            â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â”‚                                                                             â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”‚  Integration Tests (medium, API validation)                          â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ API endpoint testing                                             â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ Database operations                                              â”‚   â”‚
â”‚  â”‚  â””â”€â”€ Service interactions                                             â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â”‚                                                                             â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”‚
â”‚  â”‚  Unit Tests (many, fast feedback)                                     â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ Service layer functions                                          â”‚   â”‚
â”‚  â”‚  â”œâ”€â”€ Utility functions                                                â”‚   â”‚
â”‚  â”‚  â””â”€â”€ Model validation                                                 â”‚   â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”‚
â”‚                                                                             â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Backend Testing

```
// Example: Service unit test
func TestSiteService_Create(t *testing.T) {
    db := setupTestDB(t)
    service := NewSiteService(db)

    // Test successful creation
    site, err := service.Create(&CreateSiteRequest{
        Domain:   "example.com",
        RootPath: "/var/www/example",
        PHPVer:   "8.2",
        WebServ:  "nginx",
    })

    require.NoError(t, err)
    require.NotNil(t, site)
    require.Equal(t, "example.com", site.Domain)

    // Test duplicate domain
    _, err = service.Create(&CreateSiteRequest{
        Domain: "example.com",
    })
    require.Error(t, err)
    require.Equal(t, ErrDomainExists, err)
}

// Example: Integration test
func TestDockerAPI_ListContainers(t *testing.T) {
    if testing.Short() {
        t.Skip("Skipping integration test")
    }

    // Requires running Docker daemon
    service := NewDockerService()
    containers, err := service.ListContainers()

    require.NoError(t, err)
    require.NotNil(t, containers)
}
```

Frontend Testing

```
// Example: Component test with React Testing Library
describe('SiteCard', () => {
  it('renders site information correctly', () => {
    const site = {
      id: 1,
      domain: 'example.com',
      php_version: '8.2',
      status: 'running'
    };

    render(<SiteCard site={site} onEdit={jest.fn()} />);

    expect(screen.getByText('example.com')).toBeInTheDocument();
    expect(screen.getByText('PHP 8.2')).toBeInTheDocument();
  });

  it('calls onEdit when edit button is clicked', () => {
    const onEdit = jest.fn();
    render(<SiteCard site={site} onEdit={onEdit} />);

    fireEvent.click(screen.getByRole('button', { name: /edit/i }));

    expect(onEdit).toHaveBeenCalledWith(1);
  });
});

// Example: Hook test
describe('useMetrics', () => {
  it('fetches and updates metrics', async () => {
    const { result } = renderHook(() => useMetrics());

    // Initial state
    expect(result.current.loading).toBe(true);

    // Wait for data
    await waitFor(() => {
      expect(result.current.loading).toBe(false);
      expect(result.current.metrics).toBeDefined();
    });
  });
});
```

Slide 18: Monitoring & Observability

System Monitoring

Metrics Collection

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”,                      Real-Time Monitoring System                             â”,
â”,                                                                             â”,
â”,   â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”,
â”,   â”,  Data Sources                                                            â”, â”,
â”,   â”,  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â” â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”   â”, â”,
â”,   â”,  â”,  gopsutil â”, â”, /proc/   â”, â”,  systemctlâ”, â”,  docker   â”, â”,  nginx   â”,   â”, â”,
â”,   â”,  â”,  (CPU,RAM)â”, â”,  stats   â”, â”,  (services)â”, â”,  (stats)  â”, â”,  (logs)  â”,   â”, â”,
â”,   â”,  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜   â”, â”,
â”,   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”,
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â”‚                                    â–¼                                         â”‚
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â”‚   â”‚  Metrics Collector Service                                            â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
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â”‚   â”‚  â”‚   CPU/Mem   â”‚ â”‚    Disk     â”‚ â”‚   Network   â”‚                      â”‚ â”‚
â”‚   â”‚  â”‚  Collector  â”‚ â”‚  Collector  â”‚ â”‚  Collector  â”‚                      â”‚ â”‚
â”‚   â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜                      â”‚ â”‚
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â”‚   â”‚  â”‚  Services   â”‚ â”‚   Docker    â”‚ â”‚    Custom   â”‚                      â”‚ â”‚
â”‚   â”‚  â”‚  Monitor    â”‚ â”‚   Monitor   â”‚ â”‚   Metrics   â”‚                      â”‚ â”‚
â”‚   â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜                      â”‚ â”‚
â”‚   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”‚
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â”‚   â”‚  WebSocket Broadcast (2-second interval)                              â”‚ â”‚
â”‚   â”‚                                                                       â”‚ â”‚
â”‚   â”‚  Clients: Dashboard â€¢ Monitor Page â€¢ External Systems                 â”‚ â”‚
â”‚   â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜ â”‚
â”‚                                                                             â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Metrics API

```go
// System metrics endpoint
type SystemMetrics struct {
    Timestamp time.Time  `json:"timestamp"`

    CPU       CPUMetrics  `json:"cpu"`
    Memory    MemMetrics  `json:"memory"`
    Disk      DiskMetrics `json:"disk"`
    Network   NetMetrics  `json:"network"`
    Services  []ServiceStatus `json:"services"`
    Uptime    time.Duration   `json:"uptime"`
}

type CPUMetrics struct {
    Percent    float64   `json:"percent"`
    User       float64   `json:"user"`
    System     float64   `json:"system"`
    Idle       float64   `json:"idle"`
    CoreCount  int       `json:"core_count"`
    PerCore    []float64 `json:"per_core"`
}

type MemMetrics struct {
    Total       uint64  `json:"total"`        // bytes
    Used        uint64  `json:"used"`
    Available   uint64  `json:"available"`
    Cached      uint64  `json:"cached"`
    UsedPercent float64 `json:"used_percent"`
}

type DiskMetrics struct {
    Partitions []Partition `json:"partitions"`
    IOStats    IOStats      `json:"io_stats"`
}

type NetMetrics struct {
    BytesSent   uint64    `json:"bytes_sent"`
    BytesRecv   uint64    `json:"bytes_recv"`
    PacketsSent uint64    `json:"packets_sent"`
    PacketsRecv uint64    `json:"packets_recv"`
    ErrIn      uint64    `json:"err_in"`
    ErrOut     uint64    `json:"err_out"`
    PerInterface []IFaceStats `json:"per_interface"`
}
```

Slide 19: Error Handling & Logging

Logging & Error Management

Structured Logging

```
// zerolog configuration
import "github.com/rs/zerolog"

func initLogger() zerolog.Logger {
    // Console writer with color
    output := zerolog.ConsoleWriter{Out: os.Stdout, TimeFormat: time.RFC3339}

    logger := zerolog.New(output).
        Level(zerolog.InfoLevel).
        With().
        Timestamp().
        Caller().
        Str("service", "itdlupanel").
        Logger()

    return logger
}

// Usage with structured logging
log.Info().
    Str("user", username).
    Str("action", "login").
    Str("ip", clientIP).
    Msg("User logged in")

log.Error().
    Err(err).
    Str("site_id", "123").
    Str("operation", "backup").
    Msg("Backup failed")
```

Error Types

```
// Custom error types
type AppError struct {
    Code    string `json:"code"`
    Message string `json:"message"`
    Details string `json:"details,omitempty"`
    Status  int    `json:"-"`
}

var (
    ErrNotFound = &AppError{
        Code:    "NOT_FOUND",
        Message: "Resource not found",
        Status:  404,
    }

    ErrUnauthorized = &AppError{
        Code:    "UNAUTHORIZED",
        Message: "Authentication required",
        Status:  401,
    }

    ErrForbidden = &AppError{
        Code:    "FORBIDDEN",
        Message: "Permission denied",
        Status:  403,
    }

    ErrValidation = &AppError{
        Code:    "VALIDATION_ERROR",
        Message: "Invalid request data",
        Status:  400,
    }

    ErrInternal = &AppError{
        Code:    "INTERNAL_ERROR",
        Message: "Internal server error",
        Status:  500,
    }
)

// Error handling middleware
func ErrorHandler() gin.HandlerFunc {
    return func(c *gin.Context) {
        c.Next()

        if len(c.Errors) > 0 {
            err := c.Errors.Last()
            appErr, ok := err.Err.(*AppError)
            if !ok {
                appErr = ErrInternal
            }

            c.JSON(appErr.Status, gin.H{
                "error": appErr.Code,
                "message": appErr.Message,
                "details": appErr.Details,
            })
        }
    }
}
```

Audit Log Format

```
{
  "id": 12345,
  "user_id": 1,
  "username": "admin",
  "action": "UPDATE",
  "resource": "sites",
  "details": {
    "site_id": 5,
    "domain": "example.com",
    "changes": {
      "php_version": {"old": "8.1", "new": "8.2"},
      "ssl_enabled": {"old": false, "new": true}
    }
  },
  "ip_address": "192.168.1.100",
  "user_agent": "Mozilla/5.0...",
  "created_at": "2026-06-30T14:25:03Z"
}
```

Slide 20: Káº¿t Luáºn & Roadmap

Tá»•ng Káº¿t

Nhá»¯ng Äiá»ƒm Ná»•i Báºt Ká»¹ Thuáºt

```
â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”
â”‚                     Technical Highlights Summary                             â”‚
â”‚                                                                             â”‚
â”‚  â”Œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”  â”‚
â”‚  â”‚  âœ… Single Binary Deployment                                          â”‚  â”‚
â”‚  â”‚     ~60MB static binary, no dependencies, CGO-free                   â”‚  â”‚
â”‚  â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤  â”‚
â”‚  â”‚  âœ… 11-Layer Security Stack                                           â”‚  â”‚
â”‚  â”‚     JWT, TOTP 2FA, RBAC, CSRF, Rate Limiting, Audit Logs             â”‚  â”‚
â”‚  â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤  â”‚
â”‚  â”‚  âœ… Real-Time WebSocket                                               â”‚  â”‚
â”‚  â”‚     Terminal (PTY) + Metrics (2s interval)                           â”‚  â”‚
â”‚  â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤  â”‚
â”‚  â”‚  âœ… 46+ Docker Endpoints                                              â”‚  â”‚
â”‚  â”‚     Full container lifecycle + Compose + 38 OneClick apps           â”‚  â”‚
â”‚  â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤  â”‚
â”‚  â”‚  âœ… Multi-Storage Backups                                             â”‚  â”‚
```

```
â”‚  â”‚     Local, FTP, S3, Google Drive, OneDrive + Encryption              â”‚  â”‚
â”‚  â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¤  â”‚
â”‚  â”‚  âœ… 43K+ Lines of Production Code                                    â”‚  â”‚
â”‚  â”‚     Well-structured, tested, maintainable                            â”‚  â”‚
â”‚  â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜  â”‚
â”‚                                                                             â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”˜
```

Phase 1: Completed âœ…

- [x] Core panel functionality (112 Go files, 43K LOC)
- [x] Docker management (46+ endpoints, 38 OneClick apps)
- [x] WebSocket terminal (xterm.js â†” PTY)
- [x] SSL certificates (Let's Encrypt auto-renewal)
- [x] Multi-language (EN/VI)
- [x] RBAC with 3 roles
- [x] TOTP 2FA
- [x] Incremental backups
- [x] File manager with Monaco Editor

Phase 2: In Progress ðŸš§

- [ ] Kubernetes cluster management
- [ ] Advanced monitoring (Prometheus/Grafana)
- [ ] Plugin marketplace
- [ ] API documentation (OpenAPI/Swagger)
- [ ] WAF (Coraza + HAProxy)

Phase 3: Planned ðŸ“‹

- [ ] Multi-server management
- [ ] Mobile app (React Native)
- [ ] AI-powered suggestions
- [ ] GitOps integration

LiÃªn Há»‡

- **Website**: https://github.com/itdlu/ITDLUPanel
- **Email**: support@itdlu.edu.vn
- **Docs**: https://docs.itdlu.edu.vn

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